

Lecture 6: Differential Equations as Mathematical Models & Linear Models (Explained)

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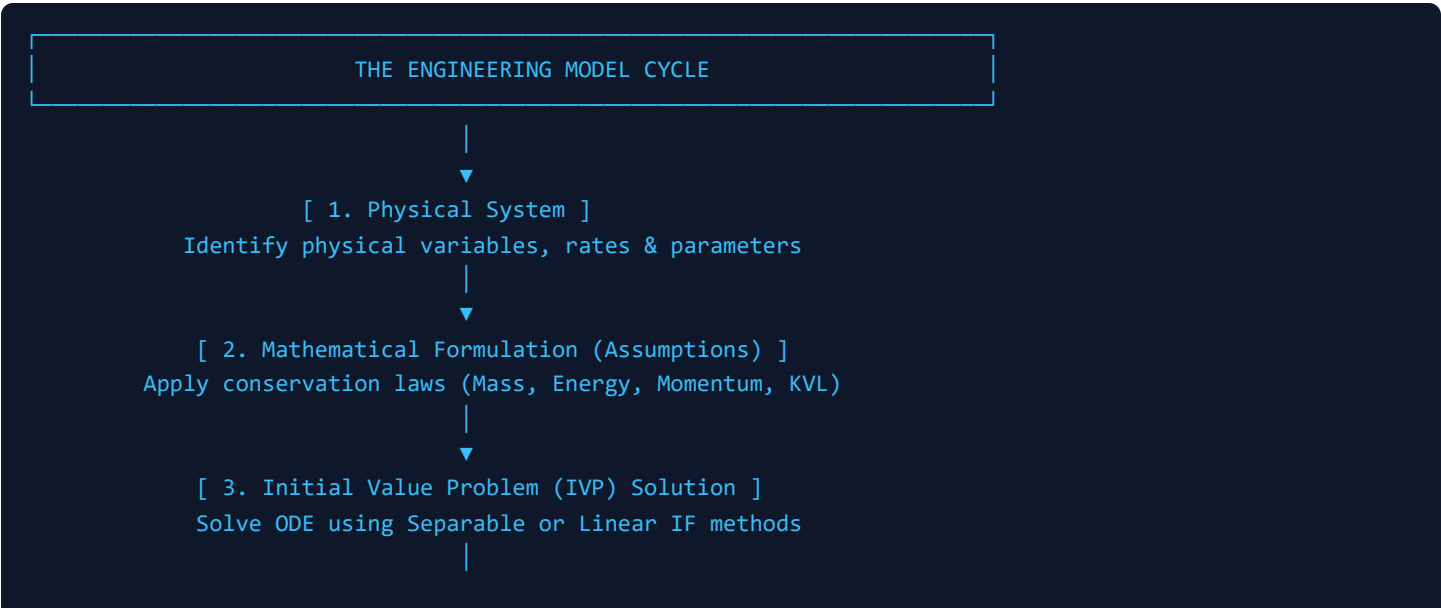
Executive Overview & Lecture Roadmap

Lecture 6 marks the fundamental bridge between purely formal ODE integration techniques and real-world engineering systems. When describing physical phenomena, empirical principles and conservation laws are formulated as **rates of change** ($\frac{dy}{dt}$), naturally producing differential equations.

LECTURE 6 LINEAR MODELING TAXONOMY (ZILL 2.7)		
Phenomenon	Governing Differential Eq.	Solution Method
Population Growth	$dP/dt = k P$	Separable
Radioactive Decay	$dA/dt = k A \quad (k < 0)$	Separable
Newton's Cooling	$dT/dt = k (T - T_m) \quad (k < 0)$	Separable / Linear
Single-Tank Mixture	$dA/dt = R_{in} - R_{out}$	1st-Order Linear
LR Series Circuit	$L (di/dt) + R i = E(t)$	1st-Order Linear
RC Series Circuit	$R (dq/dt) + (1/C) q = E(t)$	1st-Order Linear

1. Steps in the Mathematical Modeling Process

Building a reliable engineering model follows five systematic stages:



▼
[4. Model Predictions]
Calculate thresholds, time horizons & asymptotes

▼
[5. Verification & Experimental Validation]
Compare with empirical data; refine assumptions if needed

2. Exponential Growth & Radioactive Decay

2.1 Theoretical Derivation

- **Hypothesis:** The rate of change of a quantity is directly proportional to the amount currently present:

$$\frac{dP}{dt} \propto P \implies \frac{dP}{dt} = kP$$

where k is the constant of proportionality.

- If $k > 0$: Exponential growth (microbiology, unrestricted population, compound interest).
- If $k < 0$: Exponential decay (nuclear physics, radioisotope dating, drug pharmacokinetics).

- **Integration:**

$$\frac{dP}{P} = k, dt \implies \ln |P| = kt + C_1 \implies P(t) = P_0 e^{kt}$$

where $P_0 = P(0)$ represents the initial quantity at $t = 0$.

2.2 Lecture 6 — Example 1: Bacterial Growth

- **Problem:** A culture initially has P_0 bacteria. At $t = 1$ h, the number is measured to be $\frac{3}{2}P_0$. Assuming growth is proportional to the current population, determine the time necessary for the population to triple.
- **Step 1: Formulate the Model & General Solution:**

$$\frac{dP}{dt} = kP \implies P(t) = P_0 e^{kt}$$

- **Step 2: Determine Rate Constant k from Benchmark Data:**

$$P(1) = \frac{3}{2}P_0 \implies P_0 e^{k(1)} = \frac{3}{2}P_0 \implies e^k = 1.5$$

$$k = \ln(1.5) = \ln\left(\frac{3}{2}\right) \approx 0.405465 \text{ h}^{-1}$$

- **Step 3: Solve for Tripling Condition $P(t) = 3P_0$:**

$$P_0 e^{kt} = 3P_0 \implies e^{kt} = 3 \implies kt = \ln 3$$

$$t = \frac{\ln 3}{k} = \frac{\ln 3}{\ln(1.5)} = \frac{1.098612}{0.405465} \approx 2.7095 \text{ hours}$$

- **Step 4: Engineering Conversion:**

$$t \approx 2 \text{ hours and } (0.7095 \times 60) \text{ min} \approx 2 \text{ h } 43 \text{ min}$$

3. Newton's Law of Cooling and Warming

3.1 Physical Principles & Governing Equation

Newton's empirical law states that the rate of change of temperature $\frac{dT}{dt}$ of a body is proportional to the instantaneous temperature difference between the body $T(t)$ and the surrounding medium T_m :

$$\frac{dT}{dt} = k(T - T_m)$$

NEWTONIAN COOLING DYNAMICS

- If $T > T_m$ (Object hotter than room):
($T - T_m > 0$). Since object cools ($dT/dt < 0$), k MUST BE NEGATIVE.
- If $T < T_m$ (Object colder than room):
($T - T_m < 0$). Object warms up ($dT/dt > 0$), confirming $k < 0$.
- Universal Rule: In all natural heat transfer, $k < 0$.

3.2 Analytic Solution

$$\frac{dT}{T - T_m} = k, dt \implies \ln |T - T_m| = kt + C_1 \implies T(t) - T_m = ce^{kt}$$

$$T(t) = T_m + ce^{kt}$$

Using the initial temperature $T(0) = T_0 \implies c = T_0 - T_m$:

$$T(t) = T_m + (T_0 - T_m)e^{kt}$$

As $t \rightarrow \infty$, since $k < 0$, $e^{kt} \rightarrow 0$, proving that $\lim_{t \rightarrow \infty} T(t) = T_m$.

3.3 Lecture 6 — Example 2: Cooling of a Cake

- **Problem:** When a cake is removed from an oven, its temperature is 300°F . Three minutes later, its temperature is 200°F . The room temperature is 70°F . How long will it take for the cake to cool off to room temperature?
- **Step 1: Set Up Initial Parameters:**
 - Medium temperature: $T_m = 70^\circ\text{F}$
 - Initial temperature: $T(0) = 300^\circ\text{F}$
 - Temperature difference: $T_0 - T_m = 300 - 70 = 230^\circ\text{F}$
 - Governing formula: $T(t) = 70 + 230e^{kt}$
- **Step 2: Solve for Heat Dissipation Constant k :**

$$T(3) = 200 \implies 70 + 230e^{3k} = 200 \implies 230e^{3k} = 130$$

$$e^{3k} = \frac{130}{230} = \frac{13}{23} \implies 3k = \ln\left(\frac{13}{23}\right)$$

$$k = \frac{1}{3} \ln\left(\frac{13}{23}\right) \approx -0.19018 \text{ min}^{-1}$$

- **Step 3: Rigorous Mathematical vs. Engineering Resolution:**

- **Pure Mathematical Limit:**

$$T(t) = 70 + 230e^{kt} = 70 \implies 230e^{kt} = 0 \implies e^{kt} = 0$$

Since $e^{kt} > 0$ for all finite t , the cake mathematically reaches exactly 70°F only as $t \rightarrow \infty$.

- **Practical Thermal Equilibrium (Engineering Threshold):** In physical thermodynamics, thermal equilibrium is achieved when the temperature difference is imperceptible ($< 0.5^\circ\text{F}$):

$$70 + 230e^{kt} = 70.5 \implies e^{kt} = \frac{0.5}{230} = \frac{1}{460}$$

$$t = \frac{\ln(1/460)}{-0.19018} = \frac{-6.13123}{-0.19018} \approx 32.24 \text{ minutes}$$

4. Tank Mixture Dynamics

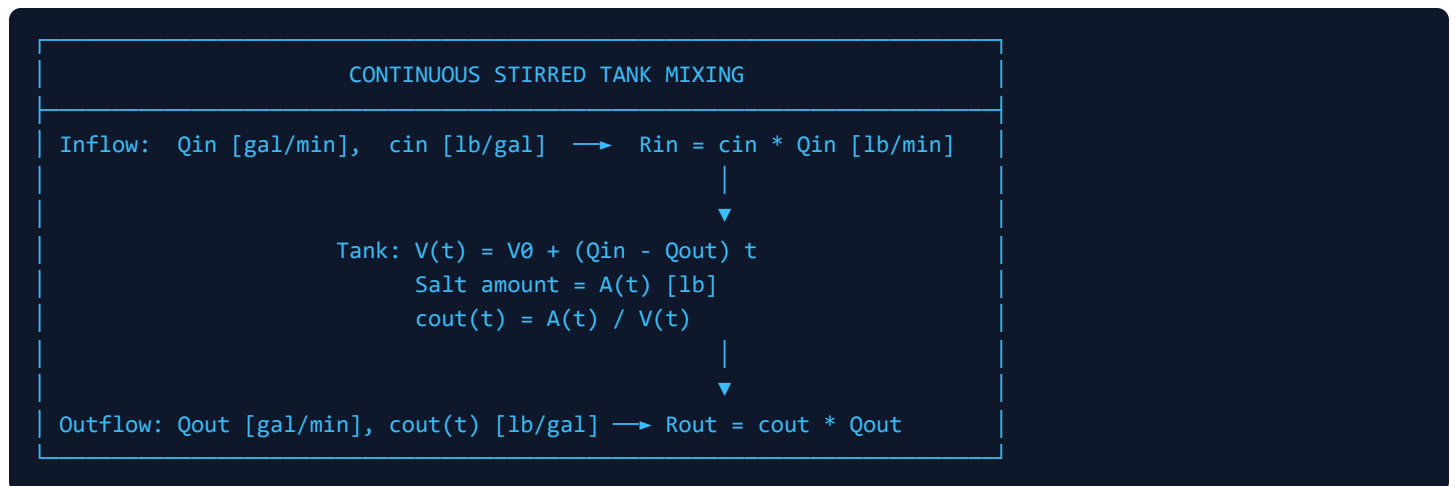
4.1 Conservation of Mass & General Formulation

The rate of accumulation of solute $A(t)$ (salt, chemical, pollutant) in a thoroughly mixed liquid tank equals input rate minus output rate:

$$\frac{dA}{dt} = R_{in} - R_{out} = c_{in}Q_{in} - c_{out}(t)Q_{out}$$

Where:

- $A(t)$ = mass of solute in the tank at time t [e.g., lb or kg]
- Q_{in}, Q_{out} = volumetric flow rates in and out [e.g., gal/min or L/min]
- c_{in} = concentration of incoming brine [e.g., lb/gal or kg/L]
- $V(t) = V_0 + (Q_{in} - Q_{out})t$ = instantaneous liquid volume in tank
- $c_{out}(t) = \frac{A(t)}{V(t)}$ = instantaneous concentration leaving tank



The universal linear ODE is:

$$\frac{dA}{dt} + \frac{Q_{out}}{V_0 + (Q_{in} - Q_{out})t} A(t) = c_{in}Q_{in}$$

4.2 Lecture 6 — Example 3: Mixture of Two Salt Solutions

- **Problem:** A large tank holds 300 gal of brine. Brine is pumped in at 3 gal/min with salt concentration 2 lb/gal, thoroughly mixed, and pumped out at the same rate of 3 gal/min. The tank initially contains 50 lb of dissolved salt. How much salt will be in the tank after a long time? Find $A(t)$.
- **Step 1: Parameter Identification:**
 - Initial volume: $V_0 = 300$ gal
 - Flow rates: $Q_{in} = 3$ gal/min, $Q_{out} = 3$ gal/min $\implies V(t) = 300$ gal (constant volume).
 - Inflow concentration: $c_{in} = 2$ lb/gal
 - Salt inflow rate: $R_{in} = c_{in}Q_{in} = (2 \text{ lb/gal})(3 \text{ gal/min}) = 6$ lb/min.
 - Salt outflow rate: $R_{out} = \left(\frac{A(t)}{300}\right)(3) = \frac{A(t)}{100}$ lb/min.
 - Initial condition: $A(0) = 50$ lb.
- **Step 2: Construct & Solve Linear ODE:**

$$\frac{dA}{dt} + \frac{1}{100}A = 6$$

- Integrating factor:

$$\mu(t) = e^{\int \frac{1}{100} dt} = e^{t/100}$$

- Multiply and integrate:

$$\frac{d}{dt} [A(t)e^{t/100}] = 6e^{t/100}$$

$$A(t)e^{t/100} = 6 \int e^{t/100} dt = 6(100)e^{t/100} + C = 600e^{t/100} + C$$

$$A(t) = 600 + Ce^{-t/100}$$

- **Step 3: Apply Initial Condition** $A(0) = 50$:

$$50 = 600 + Ce^0 \implies C = 50 - 600 = -550$$

$$A(t) = 600 - 550e^{-t/100} \text{ lb}$$

- **Step 4: Long-Term Asymptotic Behavior** ($t \rightarrow \infty$):

$$\lim_{t \rightarrow \infty} A(t) = 600 - 550(0) = 600 \text{ lb}$$

- **Physical Verification:** The steady-state amount equals total tank volume multiplied by inflow concentration:

$$A_{\infty} = V_0 \times c_{in} = (300 \text{ gal}) \times (2 \text{ lb/gal}) = 600 \text{ lb}$$

5. First-Order Series Electric Circuits

5.1 Kirchhoff's Voltage Law (KVL)

In any closed electrical loop, the impressed electromotive force $E(t)$ equals the sum of voltage drops across circuit components:

$$V_L + V_R + V_C = E(t)$$

Component	Constitutive Law	Voltage Drop
Inductor	$v_L = L (di/dt)$	$L (di/dt)$
Resistor	$v_R = R i$	$R i$
Capacitor	$v_C = (1/C) q$	$(1/C) q$ where $i = dq/dt$

5.2 LR-Series Circuit Analysis

When an inductor L and resistor R are connected in series with voltage source $E(t)$:

$$L \frac{di}{dt} + Ri = E(t) \implies \frac{di}{dt} + \frac{R}{L}i = \frac{E(t)}{L}$$

- Integrating factor: $\mu(t) = e^{\int \frac{R}{L} dt} = e^{\frac{R}{L}t}$
- For constant voltage $E(t) = E_0$:

$$i(t) = \frac{E_0}{R} + ce^{-\frac{R}{L}t}$$

If $i(0) = 0$:

$$i(t) = \frac{E_0}{R} \left(1 - e^{-\frac{R}{L}t}\right)$$

- Steady-State Current:** $i_{ss} = \frac{E_0}{R}$ (inductor acts as a short circuit).
- Transient Current:** $i_{tr}(t) = -\frac{E_0}{R} e^{-\frac{R}{L}t}$.
- Inductive Time Constant:** $\tau = \frac{L}{R}$. At $t = \tau$, $i(\tau) = 0.632 i_{ss}$.

5.3 Lecture 6 — Example 4: LR-Series Circuit

- Problem:** A 12 V battery is connected to an LR series circuit with inductance $L = 0.5$ H and resistance $R = 10 \Omega$. Determine current $i(t)$ if initial current is zero.
- Step 1: Set Up Governing Equation:**

$$0.5 \frac{di}{dt} + 10i = 12 \implies \frac{di}{dt} + 20i = 24$$

- Step 2: Solve Using Integrating Factor:**

$$\mu(t) = e^{\int 20 dt} = e^{20t}$$

$$\frac{d}{dt} [ie^{20t}] = 24e^{20t} \implies i(t)e^{20t} = \frac{24}{20}e^{20t} + C = 1.2e^{20t} + C$$

$$i(t) = 1.2 + Ce^{-20t}$$

- Step 3: Apply Initial Condition** $i(0) = 0$:

$$0 = 1.2 + C \implies C = -1.2$$

$$i(t) = 1.2 (1 - e^{-20t}) \text{ Amperes}$$

- Step 4: Engineering Interpretation:**

- Steady-state current: $\lim_{t \rightarrow \infty} i(t) = 1.2 \text{ A}$.
- Time constant: $\tau = \frac{L}{R} = \frac{0.5}{10} = 0.05 \text{ seconds}$.
- By $t = 5\tau = 0.25 \text{ seconds}$, current reaches 99.3 of its maximum value.